FaultGraph Lab: guarded chain compilation

Research v4 | Frozen prospective comparison | AI-assisted research; human review pending

Result and scientific scope

The recorded primary outcome is **SUPPORTED_ON_FROZEN_SCOPE**: **6 of 6** provider/core clusters meet every numerical threshold; at least 3 are required. Complete certified accounting: **true**. This statement reports the frozen analysis without changing acceptance criteria or discarding failures.

The hypothesis requires a chain candidate warm-build ratio at most 0.80 against **both** direct-symbolic CUDD and the earlier AIG-to-CUDD compiler, together with portable-query p50 and total peak RSS ratios at most 1.10 versus direct CUDD. Every required arm must certify; missing/unknown outputs leave the primary result inconclusive. No novelty or universal superiority follows from a positive result.

Why the obligation matters

Replacing a degree-two chain by an all-active macro edge alone is unsound for all-terminal connectivity: two failed chain edges can isolate required internal vertices. The candidate also keeps A = at most one inactive chain edge as a global Boolean obligation. It combines these events on the original processor variables, preserving shared and negated predicates. Closed cycles retain A; parallel macro edges are ORed. The output is a complete decision diagram, not the reduced graph alone.

Fresh external subsystem data

90 cases cover 6 fresh archived provider cores: Abvt, Aconet, Agis, Arnes, AsnetAm, Bandcon. Their hypothetical single-link, redundant-pair and shared-controller rules use 21-60 processor variables and five fixed assignment seeds. Selection used the first eligible providers and largest biconnected cores under declared size bounds, without runtime or degree-two filtering. Physical parallel links remain distinct in the domain oracle and are OR-grouped only in the logical graph.

These are subsystem models of historical topologies with hypothetical failure policies, not measurements of whole-provider availability. Ninety model instances and thousands of states are not ninety independent networks. The declared aggregation takes the median of three policy ratios within each seed, then the median of five seed medians; bootstrap resamples six whole-provider clusters.

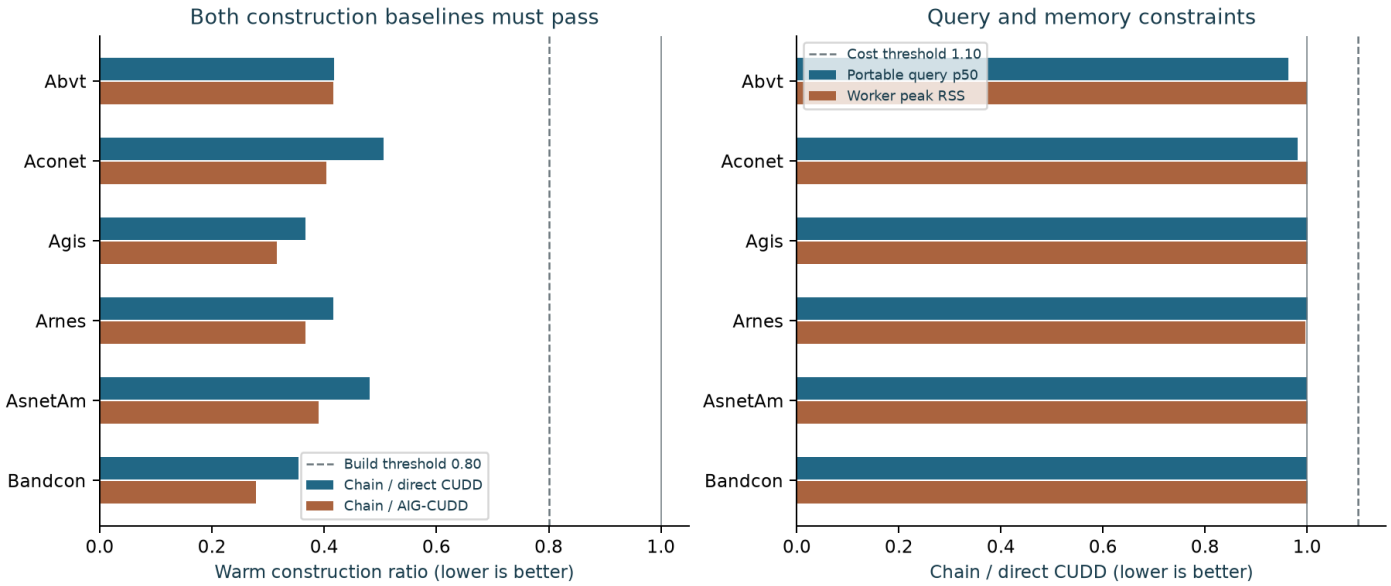
Closest prior art and attribution

Degree-two, series and polygon-to-chain reliability reductions are established. Resende (1986) explicitly treats required-terminal degree-two reduction with a global reliability factor; Satyanarayana and Wood (1985), with a later 1996 correction, are also close prior art. This implementation's correlated Boolean form and benchmark scope do not establish algorithmic novelty. Original GL modeling mathematics remains attributed to the coauthored Romankevich et al. 2026 studies.

Supplemental external-tool memory

A separate post-hoc local profile covers 90 ABC cases: 90 outputs match the primary bytes and 90 CEC checks pass. The median maximum child RSS after synthesis and CEC is 22.38 MiB (maximum 28.03 MiB). Primary worker RSS excludes this child process. These later measurements are reported separately; parent and child high-water marks are never added or called a simultaneous process-tree peak.

Primary paired measurements



Provider/core	Build/direct	Build/AIG	Query/direct	RSS/direct	Qualifies
Abvt	0.417	0.416	0.964	1.000	yes
Aconet	0.505	0.404	0.982	1.001	yes
Agis	0.366	0.315	1.000	1.001	yes
Arnes	0.417	0.367	1.001	0.997	yes
AsnetAm	0.481	0.390	1.000	1.001	yes
Bandcon	0.354	0.278	1.000	1.001	yes

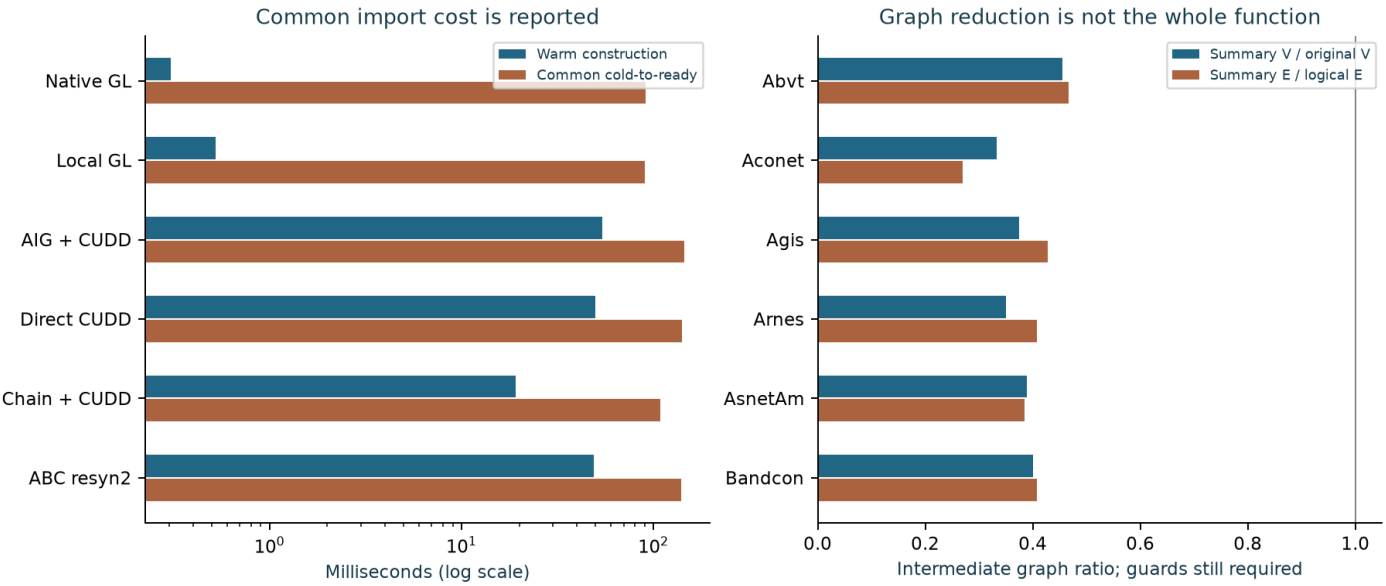
Correctness, limits and negative cases

All retained records: 540/540 planned. Status inventory: COMPLETE: 540. Recorded exact output certificates: 540; independent-oracle exhaustive comparisons: 0; supplementary sampled checks: 277,560. Sampling is not a proof. Source-rule checks, SMT equivalence, ABC CEC, raw witnesses and UNKNOWN/resource outcomes are retained separately.

Among 90 valid paired cells, candidate costs exceed the direct baseline in 0 warm-build, 41 query and 51 RSS measurements; it exceeds AIG-CUDD build cost in 0 cells. Its exported BDD has more nodes in 54 cells. 6 individual query ratios exceed 1.10 despite passing cluster medians. These descriptive counts include timing noise and are not independent family-level significance tests; no per-case non-regression guarantee follows.

Equal envelopes: 30s construction, 60s per exact query including encoding, 180s worker; CUDD configured to 256 MiB with checked 100,000 manager/export/gate limits. The manager-node high-water count is sampled between public calls; it is not an internal allocation peak. Worker RSS includes verification, with construction RSS also recorded; ABC child-process memory is excluded. Finite UNSAT applies to each encoded instance, not to every possible implementation input.

Construction, representation and reliability



Descriptive medians across certified cells: warm construction 19.14ms for the candidate, 49.71ms for direct CUDD and 54.20ms for AIG-CUDD. Common cold-to-ready medians are 108.72ms versus 140.98ms for direct CUDD. These pooled absolute costs are secondary summaries; the primary decision uses the declared paired cluster ratios.

Warm construction includes original model building, compilation and portable export; serialization, metric calculation and certificate conversion are separate. Every fresh worker imports the same CUDD/Z3 dependencies first. The common cold-to-ready cost is shown explicitly, but is not minimal native-GL startup. Each query workload has 256 uniform and 256 high-working-probability states, five repeats and 16 warm-ups. Timed workers run sequentially with deterministic arm rotation.

Candidate versus direct/AIG dynamic diagrams have maximum absolute probability disagreement **6.66e-16** across 540 paired probability settings, each comparing independent and common-cause laws (1080 numerical differences). Laws use independent original processors at p=0.8, 0.9 and 0.99; the correlated case puts all processors down with q=0.1 and otherwise uses the independent law. Derived edges and chain obligations are never treated as independent events.

Source Boolean tree operations, CUDD public apply calls, AIG gates, exported decision nodes and graph V/E are different quantities. Compactness does not imply lower runtime. Graph ratios describe the intermediate topology; global obligations remain in the measured compiled output. Their counts and BDD/apply metrics appear in diagnostics. The timing result remains bounded to this dataset and environment.

Engineering evidence and open review

Attached integrated pytest log: **284 passed**. Second-environment receipt: MATCHED_FINITE_EVIDENCE; 540 compared records; status/truth/output differences 0/0/0; source a7196baaebe4. CI run evidence. Linux also has 6/6 qualifying clusters. CI and separately delegated agent audits are engineering evidence, not independent human scientific review. Original paper reproductions and the negative/inconclusive v2/v3 studies remain preserved. Human source/closest-prior-art review, coauthor/IP decisions and public-release approval remain unsigned.

References: Romankevich et al., DOI 10.5815/ijem.2026.03.15; 10.15276/hait.09.2026.03; 10.15276/aait.09.2026.25. Satyanarayana/Wood, DOI 10.1137/0214057; correction DOI 10.1137/S0097539793257988. Resende, IEEE Trans. Reliability R-35(1), 24-29 (1986), author-hosted polychain paper. Topology Zoo: Knight et al., DOI 10.1109/JSAC.2011.111002; source archive DOI 10.25909/30153949, CC BY4.0. Pinned sources, CUDD/ABC/Z3 versions, exact commands and evidence accompany the repository.